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Evalyn

Automated Answer Sheet Evaluator



HOW IT WORKS

Two PDFs in, one result out

01

Upload

Answer key PDF and student sheet PDF.

02

Extract

Typed text is read and split into numbered questions.

03

Grade

Each answer is scored by by its question type.

04

Report

Marks, percentage and Pass/Fail.

GRADING

Grading adapts to each question type

Type	How it is marked
MCQ, True/False	Exact match with the key
Fill in the blank	Exact match, tolerating a one-letter typo
Numeric	Within rounding tolerance of the key value
Short, Long	Keyword coverage, flagged for human review

CODE WALKTHROUGH

Python code structure

01

Flask routes

/ serves the page. **/api/evaluate** validates the uploads and returns the result as JSON.

02

Grading engine

Plain Python functions extract text, parse the key, grade each answer and total the marks.

03

Command line

main() uses argparse: pass two PDF paths for a printed report, or none to to start the site.

CODE WALKTHROUGH

Python libraries used

Library	Role in Evalyn
flask	Web server, routing, request.files, jsonify, send_from_directory
pypdf	PdfReader extracts selectable text; PdfReadError catches bad files
re	Regular expressions find questions, marks, types, answers and keywords
decimal	Decimal compares numeric answers exactly, with no float drift
argparse	Command-line options for the PDF paths, pass mark and port
pathlib, sys	File paths and error output

CODE WALKTHROUGH

Functions: reading the PDFs

Function	What it does
extract_pdf_text	Reads every page; rejects pages with no selectable text
split_questions	Splits text at markers like 1., Q2 or Question 3 into a number-to-text map
parse_answer_key	Pulls marks, type, Answer and Keywords out of each key block
normalize	Lowercases and strips punctuation so answers compare fairly
keywords_from	Turns text into a set of meaningful words, dropping stop words

CODE WALKTHROUGH

Functions: grading and serving

Function	What it does
first_number	Finds the first numeric value in text, handling commas and decimals
one_edit_apart	True if two words differ by at most one letter
grade_answer	Scores one answer by type: exact, typo-tolerant, numeric or keyword share
evaluate	Grades every key question, then totals marks, percentage and PASS/FAIL
evaluate_uploads	Flask route: validates input, adds feedback and review flags
main	Starts the website, or grades two PDFs from the command line

Evalyn

Answer Sheet Evaluator

Upload a typed answer key and a student's answer sheet (PDF) with selectable text to get question-wise marks, a total and a Pass/Fail result. The number of questions can vary; scanned and handwritten pages aren't supported.

Answer key PDF
Click or drop a file here

Student answer sheet PDF
Click or drop a file here

Pass mark (%)
40

Student name (optional)
Read from the sheet if left blank

Evaluate

▼ Answer key format

Add one numbered block per question; there is no fixed question count. The type is inferred from the answer and question text. You can optionally add a type tag to override it. Student answers should start with `1.`, `Q1` or similar.

Both PDFs must contain selectable typed text. Scanned and handwritten pages aren't supported.

```
Q1 [2 marks]
Question: Which organelle is the powerhouse of the cell?
Answer: B

Q2 [5 marks]
Question: What is photosynthesis?
Answer: The process by which plants make glucose from CO2 and water using sunlight.
Keywords: chlorophyll, sunlight, glucose

Q3 [10 marks]
Question: Explain the water cycle.
Answer: ...
Rubric:
- Definition 2 marks
- Process 8 marks

Q4 [2 marks] [type: blank]
Question: The green pigment in plants is _____.
Answer: chlorophyll
```

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Answer key PDF
Answer_Key.pdf

Student answer sheet PDF
Student_Answer_Sheet.pdf

Pass mark (%)
40

Student name (optional)
Read from the sheet if left blank

Evaluate

▶ Answer key format

Student: **Asha Raman**

18.18 / 25 (73%)

PASS (provisional)

Pass mark: 40%

Some written answers need human review because marks use approximate keyword matching.

Q	TYPE	MARKS	FEEDBACK
Q1	mcq	2 / 2	Correct.
Q2	tf	2 / 2	Correct.
Q3	blank	2 / 2	Correct.
Q4	numeric	4 / 4	Correct.
Q5	short	3.33 / 5	Written-answer marks use keyword matching. review
Q6	long	4.85 / 10	Written-answer marks use keyword matching. review

Print | Download JSON

THE RESULT

A clear, reviewable result

01

Scores

Question-wise marks, total, percentage and Pass/Fail against a configurable pass mark (default 40%).

02

Review flags

Written answers are marked provisional so a teacher confirms them.

03

Export

Print the report or download JSON. Runs as a local Python web app or from the command line.

Thank You

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